Supplementary material for

CLAMP: Fairness-Constrained Federated Graph Learning from Validated Sums with Private Group Statistics

Quang-Vinh Dang, Ngoc-Son-An Nguyen

S1. Full results of the main comparison

Table S1 extends the main comparison with the backbone of each method, the hard demographic-parity gap at threshold 0.5, the equal-opportunity gap, the record $\lambda^{(R)}/(\eta R)$ of the dual update for the CLAMP variants, and the number of seeds whose test gap is within the budget $\tau = 0.03$.

Table S1: Full main comparison, mean±sd over seeds, for all datasets and methods. DPD (soft) is the demographic-parity gap of the scores; DPD (hard) uses a 0.5 threshold. Cert.: $\lambda^R/(\eta R)$ for the CLAMP variants. Backbone: GCN or GAT as in the method definition, own: the baseline's own architecture. Degenerate runs (AUC near chance) are included.

Dataset	Method	Backbone	AUC	DPD (soft)	DPD (hard)	EOD	Cert.	DPD ≤ 0.03
German	FedAvg-GCN	GCN	0.74±0.05	0.087±0.077	0.092±0.085	0.082±0.067	–	3/10
German	FedAvg-GAT	GAT	0.76±0.05	0.110±0.058	0.110±0.083	0.067±0.058	–	0/10
German	FairFed	GCN	0.75±0.05	0.049±0.039	0.077±0.063	0.056±0.069	–	4/10
German	FairGNN	own	0.57±0.06	0.111±0.082	0.114±0.082	0.140±0.099	–	3/10
German	FairSIN	own	0.72±0.05	0.092±0.062	0.099±0.069	0.066±0.054	–	1/10
German	F2GNN	GAT	0.73±0.02	0.042±0.030	0.065±0.044	0.051±0.057	–	4/10
German	FairGFL	GCN	0.74±0.05	0.081±0.070	0.096±0.075	0.064±0.054	–	2/10
German	FedGraph-Fair	GCN	0.66±0.09	0.080±0.051	0.076±0.065	0.070±0.063	–	1/10
German	PUFFLE	GCN	0.49±0.08	0.011±0.007	0.082±0.087	0.089±0.087	–	10/10
German	FedFACT	GCN	0.72±0.04	0.078±0.064	0.071±0.028	0.073±0.048	–	4/10
German	FedFDP	GCN	0.53±0.09	0.014±0.021	0.071±0.082	0.065±0.085	–	9/10
German	ClipMean (no controller, last iterate)	GAT	0.73±0.06	0.088±0.069	0.110±0.083	0.085±0.077	–	1/10
German	CLAMP (last iterate)	GAT	0.73±0.07	0.050±0.046	0.052±0.055	0.077±0.041	0.0053	5/10
German	CLAMP-GCN (last iterate)	GCN	0.73±0.05	0.041±0.044	0.058±0.056	0.048±0.044	0.0049	7/10
German	CLAMP	GAT	0.72±0.06	0.058±0.044	0.073±0.060	0.089±0.053	0.0053	3/10
German	CLAMP-GCN	GCN	0.73±0.05	0.065±0.044	0.097±0.053	0.061±0.063	0.0049	3/10
German	CLAMP-Select (no controller)	GAT	0.73±0.06	0.077±0.066	0.094±0.076	0.081±0.067	–	1/10
Bail	FedAvg-GCN	GCN	0.80±0.03	0.020±0.011	0.032±0.018	0.018±0.016	–	8/10
Bail	FedAvg-GAT	GAT	0.80±0.04	0.019±0.011	0.023±0.016	0.032±0.022	–	8/10
Bail	FairFed	GCN	0.80±0.03	0.012±0.007	0.020±0.013	0.033±0.028	–	10/10
Bail	FairGNN	own	0.56±0.09	0.019±0.020	0.021±0.022	0.019±0.016	–	7/10
Bail	FairSIN	own	0.79±0.03	0.025±0.008	0.033±0.019	0.017±0.011	–	7/10
Bail	F2GNN	GAT	0.78±0.03	0.012±0.008	0.019±0.013	0.031±0.025	–	10/10

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Dataset	Method	Backbone	AUC	DPD (soft)	DPD (hard)	EOD	Cert.	DPD ≤ 0.03
Bail	FairGFL	GCN	0.78±0.03	0.015±0.006	0.021±0.012	0.027±0.018	–	10/10
Bail	FedGraph-Fair	GCN	0.76±0.03	0.010±0.007	0.018±0.014	0.032±0.020	–	10/10
Bail	PUFFLE	GCN	0.50±0.06	0.003±0.003	0.010±0.010	0.016±0.014	–	10/10
Bail	FedFACT	GCN	0.80±0.03	0.013±0.008	0.011±0.005	0.061±0.022	–	10/10
Bail	FedFDP	GCN	0.51±0.03	0.004±0.003	0.019±0.020	0.014±0.015	–	10/10
Bail	ClipMean (no controller, last iterate)	GAT	0.78±0.03	0.016±0.009	0.020±0.014	0.031±0.024	–	10/10
Bail	CLAMP (last iterate)	GAT	0.78±0.03	0.016±0.009	0.020±0.014	0.031±0.023	0.0000	10/10
Bail	CLAMP-GCN (last iterate)	GCN	0.79±0.02	0.014±0.006	0.022±0.014	0.028±0.020	0.0000	10/10
Bail	CLAMP	GAT	0.78±0.03	0.016±0.009	0.021±0.014	0.028±0.024	0.0000	10/10
Bail	CLAMP-GCN	GCN	0.79±0.02	0.015±0.006	0.022±0.015	0.028±0.022	0.0000	10/10
Bail	CLAMP-Select (no controller)	GAT	0.78±0.03	0.016±0.009	0.021±0.014	0.029±0.024	–	10/10
Credit	FedAvg-GCN	GCN	0.76±0.01	0.072±0.010	0.110±0.022	0.098±0.025	–	0/10
Credit	FedAvg-GAT	GAT	0.77±0.01	0.078±0.010	0.116±0.018	0.103±0.017	–	0/10
Credit	FairFed	GCN	0.75±0.01	0.008±0.005	0.048±0.031	0.040±0.030	–	10/10
Credit	FairGNN	own	0.67±0.09	0.100±0.055	0.099±0.055	0.085±0.047	–	2/10
Credit	FairSIN	own	0.75±0.00	0.070±0.012	0.105±0.024	0.094±0.029	–	0/10
Credit	F2GNN	GAT	0.76±0.01	0.006±0.007	0.063±0.022	0.044±0.024	–	10/10
Credit	FairGFL	GCN	0.76±0.01	0.072±0.009	0.108±0.023	0.091±0.024	–	0/10
Credit	FedGraph-Fair	GCN	0.75±0.01	0.075±0.014	0.100±0.026	0.083±0.032	–	0/10
Credit	PUFFLE	GCN	0.46±0.13	0.016±0.008	0.070±0.076	0.075±0.075	–	10/10
Credit	FedFACT	GCN	0.75±0.01	0.070±0.016	0.045±0.028	0.056±0.034	–	0/10
Credit	FedFDP	GCN	0.54±0.11	0.017±0.009	0.073±0.056	0.071±0.055	–	9/10
Credit	ClipMean (no controller, last iterate)	GAT	0.77±0.01	0.071±0.009	0.104±0.014	0.088±0.021	–	0/10
Credit	CLAMP (last iterate)	GAT	0.76±0.01	0.055±0.046	0.098±0.063	0.087±0.060	0.0073	5/10
Credit	CLAMP-GCN (last iterate)	GCN	0.75±0.01	0.046±0.031	0.080±0.045	0.075±0.049	0.0109	2/10
Credit	CLAMP	GAT	0.76±0.01	0.018±0.015	0.056±0.018	0.050±0.019	0.0081	8/10
Credit	CLAMP-GCN	GCN	0.75±0.01	0.014±0.013	0.036±0.018	0.034±0.023	0.0104	9/10
Credit	CLAMP-Select (no controller)	GAT	0.77±0.01	0.067±0.010	0.097±0.023	0.087±0.022	–	0/10
Pokec-z	FedAvg-GCN	GCN	0.62±0.07	0.038±0.010	0.037±0.017	0.063±0.032	–	1/10
Pokec-z	FedAvg-GAT	GAT	0.74±0.02	0.031±0.013	0.036±0.018	0.040±0.027	–	5/10
Pokec-z	FairFed	GCN	0.64±0.05	0.004±0.003	0.010±0.009	0.034±0.027	–	10/10
Pokec-z	FairGNN	own	0.54±0.08	0.005±0.013	0.005±0.013	0.008±0.019	–	9/10
Pokec-z	FairSIN	own	0.66±0.05	0.026±0.010	0.038±0.023	0.041±0.027	–	6/10
Pokec-z	F2GNN	GAT	0.74±0.02	0.008±0.007	0.012±0.008	0.025±0.017	–	10/10
Pokec-z	FairGFL	GCN	0.66±0.02	0.041±0.011	0.043±0.015	0.077±0.021	–	1/10
Pokec-z	FedGraph-Fair	GCN	0.64±0.03	0.039±0.013	0.041±0.017	0.072±0.025	–	2/10
Pokec-z	PUFFLE	GCN	0.55±0.11	0.008±0.009	0.022±0.022	0.026±0.024	–	9/10
Pokec-z	FedFACT	GCN	0.63±0.04	0.024±0.014	0.034±0.024	0.043±0.033	–	7/10
Pokec-z	FedFDP	GCN	0.64±0.11	0.008±0.008	0.029±0.025	0.019±0.019	–	10/10
Pokec-z	ClipMean (no controller, last iterate)	GAT	0.75±0.02	0.042±0.011	0.055±0.020	0.030±0.026	–	1/10
Pokec-z	CLAMP (last iterate)	GAT	0.75±0.02	0.027±0.004	0.028±0.008	0.029±0.030	0.0007	8/10

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Dataset	Method	Backbone	AUC	DPD (soft)	DPD (hard)	EOD	Cert.	DPD ≤ 0.03
Pokec-z	CLAMP-GCN (last iterate)	GCN	0.72±0.01	0.028±0.006	0.025±0.010	0.041±0.033	0.0012	8/10
Pokec-z	CLAMP	GAT	0.75±0.02	0.026±0.004	0.025±0.012	0.026±0.026	0.0008	9/10
Pokec-z	CLAMP-GCN	GCN	0.72±0.02	0.028±0.006	0.027±0.011	0.047±0.029	0.0013	8/10
Pokec-z	CLAMP-Select (no controller)	GAT	0.75±0.02	0.041±0.010	0.069±0.026	0.032±0.024	–	2/10
Pokec-n	FedAvg-GCN	GCN	0.62±0.06	0.031±0.018	0.025±0.022	0.036±0.037	–	4/10
Pokec-n	FedAvg-GAT	GAT	0.75±0.02	0.034±0.017	0.037±0.019	0.041±0.024	–	4/10
Pokec-n	FairFed	GCN	0.62±0.06	0.006±0.004	0.007±0.005	0.032±0.023	–	10/10
Pokec-n	FairGNN	own	0.50±0.00	0.000±0.000	0.000±0.000	0.000±0.000	–	10/10
Pokec-n	FairSIN	own	0.64±0.06	0.043±0.014	0.049±0.034	0.046±0.037	–	2/10
Pokec-n	F2GNN	GAT	0.74±0.03	0.008±0.007	0.006±0.005	0.026±0.017	–	10/10
Pokec-n	FairGFL	GCN	0.65±0.03	0.044±0.007	0.042±0.013	0.035±0.024	–	0/10
Pokec-n	FedGraph-Fair	GCN	0.59±0.05	0.024±0.015	0.018±0.012	0.020±0.021	–	7/10
Pokec-n	PUFFLE	GCN	0.57±0.11	0.010±0.006	0.038±0.025	0.028±0.022	–	10/10
Pokec-n	FedFACT	GCN	0.59±0.06	0.012±0.011	0.016±0.014	0.046±0.034	–	9/10
Pokec-n	FedFDP	GCN	0.66±0.08	0.010±0.008	0.026±0.037	0.024±0.020	–	10/10
Pokec-n	ClipMean (no controller, last iterate)	GAT	0.75±0.02	0.052±0.016	0.064±0.025	0.043±0.027	–	1/10
Pokec-n	CLAMP (last iterate)	GAT	0.75±0.02	0.028±0.012	0.027±0.015	0.024±0.013	0.0010	5/10
Pokec-n	CLAMP-GCN (last iterate)	GCN	0.71±0.02	0.028±0.009	0.026±0.012	0.027±0.021	0.0015	7/10
Pokec-n	CLAMP	GAT	0.76±0.02	0.028±0.012	0.028±0.016	0.029±0.018	0.0010	5/10
Pokec-n	CLAMP-GCN	GCN	0.71±0.02	0.027±0.008	0.026±0.012	0.022±0.017	0.0015	6/10
Pokec-n	CLAMP-Select (no controller)	GAT	0.75±0.02	0.052±0.015	0.090±0.027	0.049±0.029	–	0/10
AMiner-S	FedAvg-GCN	GCN	0.93±0.00	0.005±0.003	0.005±0.003	0.027±0.018	–	10/10
AMiner-S	FedAvg-GAT	GAT	0.94±0.00	0.007±0.004	0.007±0.003	0.017±0.011	–	10/10
AMiner-S	FairFed	GCN	0.94±0.00	0.005±0.003	0.006±0.004	0.024±0.021	–	10/10
AMiner-S	FairGNN	own	0.63±0.08	0.098±0.035	0.072±0.057	0.098±0.095	–	0/10
AMiner-S	FairSIN	own	0.92±0.01	0.005±0.003	0.005±0.003	0.023±0.020	–	10/10
AMiner-S	F2GNN	GAT	0.94±0.01	0.005±0.004	0.006±0.004	0.017±0.013	–	10/10
AMiner-S	FairGFL	GCN	0.94±0.01	0.003±0.002	0.003±0.003	0.019±0.019	–	10/10
AMiner-S	FedGraph-Fair	GCN	0.93±0.01	0.003±0.003	0.004±0.003	0.024±0.016	–	10/10
AMiner-S	PUFFLE	GCN	0.49±0.03	0.006±0.005	0.008±0.008	0.022±0.019	–	10/10
AMiner-S	FedFACT	GCN	0.92±0.01	0.008±0.004	0.004±0.003	0.028±0.016	–	10/10
AMiner-S	FedFDP	GCN	0.49±0.03	0.006±0.004	0.011±0.010	0.021±0.024	–	10/10
AMiner-S	ClipMean (no controller, last iterate)	GAT	0.94±0.01	0.005±0.004	0.004±0.005	0.013±0.009	–	10/10
AMiner-S	CLAMP (last iterate)	GAT	0.94±0.01	0.005±0.003	0.004±0.005	0.012±0.009	0.0000	10/10
AMiner-S	CLAMP-GCN (last iterate)	GCN	0.94±0.00	0.002±0.002	0.003±0.003	0.019±0.022	0.0000	10/10
AMiner-S	CLAMP	GAT	0.94±0.01	0.004±0.004	0.004±0.005	0.014±0.009	0.0000	10/10
AMiner-S	CLAMP-GCN	GCN	0.94±0.00	0.002±0.002	0.003±0.002	0.017±0.018	0.0000	10/10
AMiner-S	CLAMP-Select (no controller)	GAT	0.94±0.01	0.005±0.003	0.004±0.005	0.014±0.010	–	10/10
NBA	FedAvg-GCN	GCN	0.78±0.04	0.084±0.047	0.085±0.049	0.105±0.090	–	1/10
NBA	FedAvg-GAT	GAT	0.81±0.06	0.099±0.059	0.102±0.055	0.088±0.066	–	1/10
NBA	FairFed	GCN	0.80±0.05	0.081±0.062	0.100±0.064	0.123±0.070	–	2/10
NBA	FairGNN	own	0.77±0.08	0.097±0.044	0.100±0.041	0.137±0.117	–	1/10

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Dataset	Method	Backbone	AUC	DPD (soft)	DPD (hard)	EOD	Cert.	DPD ≤ 0.03
NBA	FairSIN	own	0.73±0.08	0.087±0.059	0.083±0.051	0.119±0.122	–	2/10
NBA	F2GNN	GAT	0.78±0.13	0.109±0.072	0.148±0.094	0.157±0.089	–	1/10
NBA	FairGFL	GCN	0.76±0.04	0.078±0.044	0.081±0.054	0.088±0.032	–	2/10
NBA	FedGraph-Fair	GCN	0.77±0.10	0.084±0.043	0.089±0.041	0.077±0.030	–	2/10
NBA	PUFFLE	GCN	0.37±0.19	0.014±0.012	0.061±0.072	0.093±0.092	–	9/10
NBA	FedFACT	GCN	0.76±0.04	0.142±0.082	0.112±0.069	0.110±0.091	–	1/10
NBA	FedFDP	GCN	0.39±0.20	0.015±0.010	0.058±0.049	0.105±0.118	–	9/10
NBA	ClipMean (no controller, last iterate)	GAT	0.78±0.08	0.093±0.068	0.107±0.083	0.111±0.079	–	3/10
NBA	CLAMP (last iterate)	GAT	0.79±0.07	0.092±0.068	0.106±0.094	0.110±0.080	0.0180	2/10
NBA	CLAMP-GCN (last iterate)	GCN	0.77±0.07	0.070±0.046	0.076±0.050	0.059±0.058	0.0188	2/10
NBA	CLAMP	GAT	0.80±0.08	0.076±0.054	0.081±0.074	0.094±0.064	0.0180	2/10
NBA	CLAMP-GCN	GCN	0.78±0.07	0.077±0.044	0.069±0.046	0.049±0.049	0.0188	1/10
NBA	CLAMP-Select (no controller)	GAT	0.79±0.09	0.061±0.051	0.084±0.060	0.099±0.072	–	4/10
Facebook	FedAvg-GCN	GCN	0.76±0.03	0.074±0.036	0.079±0.035	0.046±0.036	–	2/10
Facebook	FedAvg-GAT	GAT	0.77±0.05	0.078±0.048	0.082±0.045	0.052±0.035	–	1/10
Facebook	FairFed	GCN	0.76±0.03	0.049±0.031	0.050±0.038	0.035±0.024	–	4/10
Facebook	FairGNN	own	0.51±0.05	0.024±0.041	0.028±0.043	0.025±0.033	–	8/10
Facebook	FairSIN	own	0.74±0.05	0.070±0.033	0.075±0.033	0.047±0.025	–	2/10
Facebook	F2GNN	GAT	0.76±0.05	0.069±0.040	0.075±0.048	0.065±0.045	–	1/10
Facebook	FairGFL	GCN	0.75±0.05	0.080±0.040	0.081±0.042	0.057±0.037	–	2/10
Facebook	FedGraph-Fair	GCN	0.74±0.05	0.094±0.037	0.091±0.039	0.061±0.033	–	1/10
Facebook	PUFFLE	GCN	0.53±0.10	0.009±0.006	0.042±0.035	0.042±0.034	–	10/10
Facebook	FedFACT	GCN	0.74±0.05	0.055±0.042	0.069±0.041	0.038±0.030	–	3/10
Facebook	FedFDP	GCN	0.54±0.08	0.013±0.009	0.044±0.041	0.059±0.065	–	10/10
Facebook	ClipMean (no controller, last iterate)	GAT	0.77±0.06	0.086±0.043	0.086±0.045	0.077±0.034	–	0/10
Facebook	CLAMP (last iterate)	GAT	0.77±0.06	0.082±0.043	0.082±0.041	0.062±0.042	0.0053	1/10
Facebook	CLAMP-GCN (last iterate)	GCN	0.76±0.05	0.071±0.031	0.081±0.028	0.064±0.035	0.0060	1/10
Facebook	CLAMP	GAT	0.76±0.06	0.069±0.034	0.080±0.037	0.067±0.042	0.0053	2/10
Facebook	CLAMP-GCN	GCN	0.76±0.04	0.078±0.030	0.089±0.049	0.084±0.049	0.0060	1/10
Facebook	CLAMP-Select (no controller)	GAT	0.76±0.06	0.082±0.032	0.096±0.034	0.084±0.044	–	0/10

S2. Update attacks, all attacker counts

Table S2 reports all attacker counts $f = 0, \ldots, 4$ with standard deviations over three seeds ($K = 10$ clients).

Table S2: Full update-attack results, mean±sd over 3 seeds, K=10 clients, for every number f of Byzantine clients. †: median, trimmed mean and Krum inspect individual client updates and cannot run under secure aggregation.

Dataset	Attack	f	Aggregator	AUC	DPD
Bail	None	0	FedAvg	0.76±0.01	0.011±0.002
Bail	None	0	Median†	0.73±0.01	0.007±0.000
Bail	None	0	Trimmed mean†	0.75±0.02	0.013±0.002
Bail	None	0	Krum†	0.72±0.02	0.003±0.001
Bail	None	0	CLAMP	0.75±0.02	0.011±0.002
Bail	None	0	ClipMean	0.75±0.02	0.009±0.003
Bail	Sign-flip	1	FedAvg	0.54±0.09	0.024±0.017
Bail	Sign-flip	1	Median†	0.72±0.01	0.009±0.004
Bail	Sign-flip	1	Trimmed mean†	0.73±0.02	0.011±0.004
Bail	Sign-flip	1	Krum†	0.71±0.01	0.009±0.004
Bail	Sign-flip	1	CLAMP	0.73±0.02	0.010±0.003
Bail	Sign-flip	1	ClipMean	0.73±0.01	0.009±0.004
Bail	Sign-flip	2	FedAvg	0.38±0.03	0.020±0.012
Bail	Sign-flip	2	Median†	0.68±0.04	0.011±0.006
Bail	Sign-flip	2	Trimmed mean†	0.70±0.04	0.012±0.001
Bail	Sign-flip	2	Krum†	0.70±0.03	0.007±0.004
Bail	Sign-flip	2	CLAMP	0.69±0.04	0.010±0.009
Bail	Sign-flip	2	ClipMean	0.69±0.04	0.010±0.009
Bail	Sign-flip	3	FedAvg	0.37±0.02	0.033±0.016
Bail	Sign-flip	3	Median†	0.67±0.04	0.007±0.006
Bail	Sign-flip	3	Trimmed mean†	0.68±0.04	0.009±0.006
Bail	Sign-flip	3	Krum†	0.71±0.01	0.009±0.007
Bail	Sign-flip	3	CLAMP	0.68±0.04	0.009±0.010
Bail	Sign-flip	3	ClipMean	0.68±0.04	0.009±0.008
Bail	Sign-flip	4	FedAvg	0.39±0.03	0.032±0.002
Bail	Sign-flip	4	Median†	0.63±0.04	0.012±0.006
Bail	Sign-flip	4	Trimmed mean†	0.65±0.03	0.023±0.011
Bail	Sign-flip	4	Krum†	0.71±0.02	0.010±0.004
Bail	Sign-flip	4	CLAMP	0.66±0.03	0.014±0.009
Bail	Sign-flip	4	ClipMean	0.66±0.04	0.013±0.010
Bail	Scaling	1	FedAvg	0.70±0.01	0.002±0.000
Bail	Scaling	1	Median†	0.73±0.01	0.009±0.002
Bail	Scaling	1	Trimmed mean†	0.76±0.02	0.012±0.001
Bail	Scaling	1	Krum†	0.71±0.01	0.009±0.004
Bail	Scaling	1	CLAMP	0.75±0.02	0.010±0.002
Bail	Scaling	1	ClipMean	0.75±0.02	0.009±0.003
Bail	Scaling	2	FedAvg	0.49±0.04	0.011±0.005

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Dataset	Attack	f	Aggregator	AUC	DPD
Bail	Scaling	2	Median†	0.73±0.03	0.010±0.003
Bail	Scaling	2	Trimmed mean†	0.75±0.01	0.012±0.007
Bail	Scaling	2	Krum†	0.71±0.02	0.006±0.005
Bail	Scaling	2	CLAMP	0.75±0.02	0.010±0.002
Bail	Scaling	2	ClipMean	0.75±0.02	0.009±0.003
Bail	Scaling	3	FedAvg	0.49±0.07	0.002±0.001
Bail	Scaling	3	Median†	0.74±0.03	0.006±0.002
Bail	Scaling	3	Trimmed mean†	0.75±0.01	0.007±0.003
Bail	Scaling	3	Krum†	0.70±0.02	0.005±0.006
Bail	Scaling	3	CLAMP	0.75±0.02	0.010±0.002
Bail	Scaling	3	ClipMean	0.75±0.02	0.009±0.003
Bail	Scaling	4	FedAvg	0.54±0.07	0.002±0.001
Bail	Scaling	4	Median†	0.73±0.02	0.005±0.005
Bail	Scaling	4	Trimmed mean†	0.75±0.01	0.008±0.002
Bail	Scaling	4	Krum†	0.71±0.02	0.010±0.004
Bail	Scaling	4	CLAMP	0.75±0.02	0.011±0.002
Bail	Scaling	4	ClipMean	0.75±0.02	0.009±0.003
Bail	ALIE	1	FedAvg	0.75±0.01	0.014±0.000
Bail	ALIE	1	Median†	0.75±0.02	0.009±0.001
Bail	ALIE	1	Trimmed mean†	0.75±0.02	0.013±0.001
Bail	ALIE	1	Krum†	0.73±0.02	0.012±0.004
Bail	ALIE	1	CLAMP	0.74±0.02	0.012±0.002
Bail	ALIE	1	ClipMean	0.74±0.02	0.011±0.004
Bail	ALIE	2	FedAvg	0.73±0.03	0.012±0.001
Bail	ALIE	2	Median†	0.72±0.03	0.007±0.003
Bail	ALIE	2	Trimmed mean†	0.73±0.03	0.009±0.002
Bail	ALIE	2	Krum†	0.73±0.03	0.008±0.001
Bail	ALIE	2	CLAMP	0.72±0.03	0.008±0.000
Bail	ALIE	2	ClipMean	0.72±0.03	0.007±0.002
Bail	ALIE	3	FedAvg	0.73±0.02	0.007±0.002
Bail	ALIE	3	Median†	0.73±0.03	0.007±0.002
Bail	ALIE	3	Trimmed mean†	0.73±0.03	0.007±0.003
Bail	ALIE	3	Krum†	0.72±0.03	0.007±0.003
Bail	ALIE	3	CLAMP	0.73±0.02	0.007±0.000
Bail	ALIE	3	ClipMean	0.73±0.03	0.006±0.002
Bail	ALIE	4	FedAvg	0.72±0.02	0.008±0.002
Bail	ALIE	4	Median†	0.70±0.03	0.002±0.002
Bail	ALIE	4	Trimmed mean†	0.70±0.03	0.002±0.001
Bail	ALIE	4	Krum†	0.70±0.03	0.002±0.002
Bail	ALIE	4	CLAMP	0.72±0.03	0.008±0.002
Bail	ALIE	4	ClipMean	0.72±0.03	0.006±0.001
Bail	IPM	1	FedAvg	0.75±0.01	0.011±0.001

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Dataset	Attack	f	Aggregator	AUC	DPD
Bail	IPM	1	Median†	0.74±0.02	0.007±0.003
Bail	IPM	1	Trimmed mean†	0.74±0.02	0.013±0.001
Bail	IPM	1	Krum†	0.57±0.00	0.014±0.014
Bail	IPM	1	CLAMP	0.74±0.02	0.010±0.002
Bail	IPM	1	ClipMean	0.74±0.02	0.010±0.003
Bail	IPM	2	FedAvg	0.72±0.02	0.010±0.001
Bail	IPM	2	Median†	0.71±0.03	0.005±0.002
Bail	IPM	2	Trimmed mean†	0.72±0.03	0.009±0.001
Bail	IPM	2	Krum†	0.47±0.02	0.005±0.005
Bail	IPM	2	CLAMP	0.72±0.03	0.007±0.001
Bail	IPM	2	ClipMean	0.72±0.03	0.006±0.002
Bail	IPM	3	FedAvg	0.72±0.02	0.009±0.002
Bail	IPM	3	Median†	0.70±0.03	0.004±0.001
Bail	IPM	3	Trimmed mean†	0.72±0.02	0.006±0.002
Bail	IPM	3	Krum†	0.42±0.03	0.022±0.014
Bail	IPM	3	CLAMP	0.72±0.02	0.006±0.002
Bail	IPM	3	ClipMean	0.72±0.02	0.006±0.002
Bail	IPM	4	FedAvg	0.71±0.02	0.009±0.002
Bail	IPM	4	Median†	0.63±0.05	0.007±0.002
Bail	IPM	4	Trimmed mean†	0.67±0.03	0.005±0.002
Bail	IPM	4	Krum†	0.38±0.03	0.022±0.006
Bail	IPM	4	CLAMP	0.71±0.02	0.005±0.002
Bail	IPM	4	ClipMean	0.71±0.02	0.005±0.001
Bail	Fairness-poison	1	FedAvg	0.75±0.01	0.008±0.002
Bail	Fairness-poison	1	Median†	0.72±0.02	0.008±0.005
Bail	Fairness-poison	1	Trimmed mean†	0.74±0.02	0.014±0.003
Bail	Fairness-poison	1	Krum†	0.72±0.01	0.005±0.004
Bail	Fairness-poison	1	CLAMP	0.74±0.02	0.012±0.003
Bail	Fairness-poison	1	ClipMean	0.74±0.02	0.011±0.002
Bail	Fairness-poison	2	FedAvg	0.72±0.02	0.014±0.011
Bail	Fairness-poison	2	Median†	0.72±0.03	0.009±0.006
Bail	Fairness-poison	2	Trimmed mean†	0.73±0.03	0.010±0.009
Bail	Fairness-poison	2	Krum†	0.71±0.01	0.009±0.001
Bail	Fairness-poison	2	CLAMP	0.72±0.03	0.011±0.008
Bail	Fairness-poison	2	ClipMean	0.72±0.03	0.010±0.008
Bail	Fairness-poison	3	FedAvg	0.71±0.03	0.018±0.005
Bail	Fairness-poison	3	Median†	0.72±0.03	0.010±0.009
Bail	Fairness-poison	3	Trimmed mean†	0.73±0.02	0.012±0.009
Bail	Fairness-poison	3	Krum†	0.71±0.02	0.005±0.004
Bail	Fairness-poison	3	CLAMP	0.72±0.02	0.009±0.009
Bail	Fairness-poison	3	ClipMean	0.72±0.02	0.016±0.009
Bail	Fairness-poison	4	FedAvg	0.70±0.03	0.021±0.008

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Dataset	Attack	f	Aggregator	AUC	DPD
Bail	Fairness-poison	4	Median†	0.71±0.04	0.013±0.008
Bail	Fairness-poison	4	Trimmed mean†	0.72±0.02	0.014±0.003
Bail	Fairness-poison	4	Krum†	0.71±0.00	0.009±0.007
Bail	Fairness-poison	4	CLAMP	0.72±0.02	0.014±0.001
Bail	Fairness-poison	4	ClipMean	0.71±0.02	0.023±0.009
Pokec-z	None	0	FedAvg	0.75±0.01	0.035±0.010
Pokec-z	None	0	Median†	0.75±0.01	0.033±0.009
Pokec-z	None	0	Trimmed mean†	0.74±0.01	0.035±0.007
Pokec-z	None	0	Krum†	0.74±0.02	0.033±0.014
Pokec-z	None	0	CLAMP	0.76±0.00	0.028±0.016
Pokec-z	None	0	ClipMean	0.76±0.00	0.045±0.007
Pokec-z	Sign-flip	1	FedAvg	0.61±0.15	0.028±0.021
Pokec-z	Sign-flip	1	Median†	0.74±0.01	0.032±0.007
Pokec-z	Sign-flip	1	Trimmed mean†	0.73±0.01	0.031±0.018
Pokec-z	Sign-flip	1	Krum†	0.74±0.01	0.034±0.013
Pokec-z	Sign-flip	1	CLAMP	0.76±0.01	0.025±0.016
Pokec-z	Sign-flip	1	ClipMean	0.76±0.01	0.041±0.009
Pokec-z	Sign-flip	2	FedAvg	0.50±0.01	0.001±0.002
Pokec-z	Sign-flip	2	Median†	0.74±0.01	0.034±0.010
Pokec-z	Sign-flip	2	Trimmed mean†	0.73±0.03	0.019±0.015
Pokec-z	Sign-flip	2	Krum†	0.74±0.02	0.029±0.007
Pokec-z	Sign-flip	2	CLAMP	0.75±0.02	0.022±0.013
Pokec-z	Sign-flip	2	ClipMean	0.75±0.02	0.032±0.014
Pokec-z	Sign-flip	3	FedAvg	0.50±0.01	0.000±0.001
Pokec-z	Sign-flip	3	Median†	0.72±0.03	0.020±0.005
Pokec-z	Sign-flip	3	Trimmed mean†	0.71±0.04	0.023±0.030
Pokec-z	Sign-flip	3	Krum†	0.72±0.02	0.023±0.007
Pokec-z	Sign-flip	3	CLAMP	0.74±0.03	0.020±0.012
Pokec-z	Sign-flip	3	ClipMean	0.74±0.03	0.023±0.015
Pokec-z	Sign-flip	4	FedAvg	0.49±0.01	0.001±0.001
Pokec-z	Sign-flip	4	Median†	0.65±0.06	0.015±0.009
Pokec-z	Sign-flip	4	Trimmed mean†	0.68±0.08	0.022±0.022
Pokec-z	Sign-flip	4	Krum†	0.72±0.01	0.033±0.009
Pokec-z	Sign-flip	4	CLAMP	0.71±0.05	0.017±0.013
Pokec-z	Sign-flip	4	ClipMean	0.71±0.05	0.016±0.011
Pokec-z	Scaling	1	FedAvg	0.72±0.01	0.033±0.025
Pokec-z	Scaling	1	Median†	0.75±0.00	0.037±0.011
Pokec-z	Scaling	1	Trimmed mean†	0.74±0.01	0.031±0.007
Pokec-z	Scaling	1	Krum†	0.75±0.02	0.036±0.013
Pokec-z	Scaling	1	CLAMP	0.76±0.00	0.028±0.016
Pokec-z	Scaling	1	ClipMean	0.76±0.00	0.045±0.008
Pokec-z	Scaling	2	FedAvg	0.64±0.01	0.029±0.025

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Dataset	Attack	f	Aggregator	AUC	DPD
Pokec-z	Scaling	2	Median†	0.76±0.01	0.033±0.003
Pokec-z	Scaling	2	Trimmed mean†	0.71±0.01	0.030±0.007
Pokec-z	Scaling	2	Krum†	0.73±0.01	0.034±0.004
Pokec-z	Scaling	2	CLAMP	0.76±0.00	0.027±0.017
Pokec-z	Scaling	2	ClipMean	0.76±0.00	0.045±0.007
Pokec-z	Scaling	3	FedAvg	0.65±0.01	0.034±0.010
Pokec-z	Scaling	3	Median†	0.75±0.02	0.035±0.012
Pokec-z	Scaling	3	Trimmed mean†	0.73±0.02	0.032±0.011
Pokec-z	Scaling	3	Krum†	0.72±0.02	0.026±0.001
Pokec-z	Scaling	3	CLAMP	0.76±0.00	0.028±0.016
Pokec-z	Scaling	3	ClipMean	0.76±0.00	0.046±0.007
Pokec-z	Scaling	4	FedAvg	0.57±0.13	0.031±0.025
Pokec-z	Scaling	4	Median†	0.76±0.01	0.036±0.014
Pokec-z	Scaling	4	Trimmed mean†	0.74±0.02	0.038±0.011
Pokec-z	Scaling	4	Krum†	0.71±0.03	0.031±0.003
Pokec-z	Scaling	4	CLAMP	0.76±0.00	0.028±0.016
Pokec-z	Scaling	4	ClipMean	0.76±0.00	0.045±0.007
Pokec-z	ALIE	1	FedAvg	0.75±0.01	0.038±0.008
Pokec-z	ALIE	1	Median†	0.75±0.00	0.034±0.015
Pokec-z	ALIE	1	Trimmed mean†	0.74±0.01	0.035±0.010
Pokec-z	ALIE	1	Krum†	0.74±0.01	0.040±0.013
Pokec-z	ALIE	1	CLAMP	0.76±0.00	0.026±0.015
Pokec-z	ALIE	1	ClipMean	0.76±0.00	0.044±0.009
Pokec-z	ALIE	2	FedAvg	0.74±0.00	0.032±0.012
Pokec-z	ALIE	2	Median†	0.74±0.01	0.032±0.013
Pokec-z	ALIE	2	Trimmed mean†	0.72±0.01	0.035±0.015
Pokec-z	ALIE	2	Krum†	0.74±0.01	0.034±0.010
Pokec-z	ALIE	2	CLAMP	0.76±0.01	0.024±0.018
Pokec-z	ALIE	2	ClipMean	0.75±0.00	0.039±0.010
Pokec-z	ALIE	3	FedAvg	0.74±0.01	0.032±0.008
Pokec-z	ALIE	3	Median†	0.75±0.01	0.031±0.006
Pokec-z	ALIE	3	Trimmed mean†	0.75±0.02	0.034±0.012
Pokec-z	ALIE	3	Krum†	0.75±0.01	0.034±0.008
Pokec-z	ALIE	3	CLAMP	0.75±0.01	0.030±0.009
Pokec-z	ALIE	3	ClipMean	0.76±0.00	0.037±0.007
Pokec-z	ALIE	4	FedAvg	0.75±0.02	0.031±0.009
Pokec-z	ALIE	4	Median†	0.72±0.01	0.029±0.010
Pokec-z	ALIE	4	Trimmed mean†	0.73±0.02	0.024±0.008
Pokec-z	ALIE	4	Krum†	0.74±0.01	0.017±0.012
Pokec-z	ALIE	4	CLAMP	0.75±0.01	0.027±0.008
Pokec-z	ALIE	4	ClipMean	0.75±0.01	0.038±0.007
Pokec-z	IPM	1	FedAvg	0.75±0.01	0.037±0.011

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Dataset	Attack	f	Aggregator	AUC	DPD
Pokec-z	IPM	1	Median†	0.76±0.00	0.037±0.008
Pokec-z	IPM	1	Trimmed mean†	0.75±0.00	0.037±0.009
Pokec-z	IPM	1	Krum†	0.69±0.08	0.017±0.003
Pokec-z	IPM	1	CLAMP	0.77±0.00	0.028±0.015
Pokec-z	IPM	1	ClipMean	0.77±0.00	0.045±0.008
Pokec-z	IPM	2	FedAvg	0.76±0.01	0.039±0.010
Pokec-z	IPM	2	Median†	0.76±0.00	0.040±0.011
Pokec-z	IPM	2	Trimmed mean†	0.75±0.00	0.039±0.010
Pokec-z	IPM	2	Krum†	0.40±0.13	0.009±0.011
Pokec-z	IPM	2	CLAMP	0.77±0.00	0.028±0.014
Pokec-z	IPM	2	ClipMean	0.77±0.00	0.046±0.009
Pokec-z	IPM	3	FedAvg	0.76±0.02	0.032±0.001
Pokec-z	IPM	3	Median†	0.77±0.01	0.035±0.008
Pokec-z	IPM	3	Trimmed mean†	0.76±0.01	0.043±0.007
Pokec-z	IPM	3	Krum†	0.35±0.04	0.004±0.002
Pokec-z	IPM	3	CLAMP	0.78±0.00	0.030±0.011
Pokec-z	IPM	3	ClipMean	0.78±0.00	0.045±0.008
Pokec-z	IPM	4	FedAvg	0.75±0.01	0.035±0.004
Pokec-z	IPM	4	Median†	0.75±0.03	0.014±0.008
Pokec-z	IPM	4	Trimmed mean†	0.77±0.01	0.027±0.004
Pokec-z	IPM	4	Krum†	0.37±0.06	0.009±0.010
Pokec-z	IPM	4	CLAMP	0.76±0.02	0.026±0.005
Pokec-z	IPM	4	ClipMean	0.76±0.02	0.026±0.005
Pokec-z	Fairness-poison	1	FedAvg	0.75±0.00	0.062±0.015
Pokec-z	Fairness-poison	1	Median†	0.74±0.02	0.046±0.027
Pokec-z	Fairness-poison	1	Trimmed mean†	0.73±0.01	0.047±0.033
Pokec-z	Fairness-poison	1	Krum†	0.74±0.01	0.035±0.002
Pokec-z	Fairness-poison	1	CLAMP	0.76±0.00	0.025±0.016
Pokec-z	Fairness-poison	1	ClipMean	0.76±0.00	0.065±0.005
Pokec-z	Fairness-poison	2	FedAvg	0.71±0.03	0.124±0.027
Pokec-z	Fairness-poison	2	Median†	0.72±0.00	0.108±0.025
Pokec-z	Fairness-poison	2	Trimmed mean†	0.70±0.01	0.146±0.036
Pokec-z	Fairness-poison	2	Krum†	0.72±0.02	0.034±0.011
Pokec-z	Fairness-poison	2	CLAMP	0.72±0.02	0.037±0.006
Pokec-z	Fairness-poison	2	ClipMean	0.75±0.00	0.137±0.017
Pokec-z	Fairness-poison	3	FedAvg	0.70±0.02	0.137±0.057
Pokec-z	Fairness-poison	3	Median†	0.72±0.01	0.148±0.036
Pokec-z	Fairness-poison	3	Trimmed mean†	0.69±0.01	0.186±0.042
Pokec-z	Fairness-poison	3	Krum†	0.73±0.03	0.035±0.012
Pokec-z	Fairness-poison	3	CLAMP	0.71±0.03	0.049±0.018
Pokec-z	Fairness-poison	3	ClipMean	0.74±0.01	0.172±0.021
Pokec-z	Fairness-poison	4	FedAvg	0.69±0.03	0.137±0.068

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Dataset	Attack	f	Aggregator	AUC	DPD
Pokec-z	Fairness-poison	4	Median†	0.72±0.02	0.111±0.031
Pokec-z	Fairness-poison	4	Trimmed mean†	0.71±0.01	0.201±0.061
Pokec-z	Fairness-poison	4	Krum†	0.71±0.04	0.086±0.099
Pokec-z	Fairness-poison	4	CLAMP	0.71±0.03	0.047±0.042
Pokec-z	Fairness-poison	4	ClipMean	0.73±0.00	0.125±0.103

S3. Privacy budget and release period on the small graphs and Bail

S4. Privacy budget with a binding budget on Bail

S5. Ablations

Table S3: Privacy: utility and fairness against the statistics budget ε and release period P. $\varepsilon = \infty$: statistics noise off, default period (P=10 for German, NBA and Facebook, P=1 for Bail). mean±sd over 5 seeds. Fallback: number of seeds with no eligible feasible release after the warm-up, so the last model is reported.

ε	P	German AUC	German DPD	fallback	Bail AUC	Bail DPD	fallback
1	1	0.73±0.08	0.052±0.034	0	0.78±0.03	0.019±0.008	0
1	5	0.65±0.07	0.163±0.099	0	0.78±0.03	0.019±0.008	0
1	10	0.66±0.07	0.244±0.160	2	0.78±0.03	0.019±0.008	0
2	1	0.71±0.06	0.090±0.079	0	0.79±0.03	0.019±0.008	0
2	5	0.69±0.04	0.212±0.141	3	0.78±0.03	0.019±0.008	0
2	10	0.74±0.08	0.144±0.026	1	0.78±0.03	0.019±0.008	0
4	1	0.70±0.05	0.046±0.058	0	0.79±0.03	0.019±0.008	0
4	5	0.72±0.08	0.113±0.072	0	0.78±0.03	0.019±0.007	0
4	10	0.73±0.08	0.102±0.042	1	0.78±0.03	0.019±0.008	0
8	1	0.71±0.06	0.124±0.121	0	0.78±0.03	0.020±0.008	0
8	5	0.73±0.08	0.068±0.020	0	0.78±0.03	0.019±0.007	0
8	10	0.73±0.09	0.081±0.049	1	0.78±0.03	0.019±0.008	0
16	1	0.73±0.04	0.059±0.055	0	0.79±0.03	0.020±0.008	0
16	5	0.73±0.08	0.064±0.036	0	0.78±0.03	0.019±0.007	0
16	10	0.73±0.09	0.069±0.036	2	0.78±0.03	0.019±0.008	0
∞	–	0.73±0.08	0.058±0.040	3	0.79±0.03	0.020±0.008	0

ε	P	NBA AUC	NBA DPD	fallback	Facebook AUC	Facebook DPD	fallback
1	1	0.79±0.10	0.065±0.063	0	0.72±0.04	0.058±0.036	0
1	5	0.82±0.07	0.072±0.057	0	0.71±0.05	0.108±0.071	2
1	10	0.74±0.14	0.077±0.069	2	0.73±0.06	0.108±0.082	3
2	1	0.79±0.11	0.096±0.018	0	0.72±0.05	0.060±0.050	0
2	5	0.74±0.06	0.103±0.041	0	0.74±0.06	0.110±0.041	2
2	10	0.74±0.10	0.121±0.087	3	0.73±0.06	0.112±0.044	3
4	1	0.76±0.12	0.101±0.056	0	0.74±0.05	0.087±0.045	0
4	5	0.77±0.10	0.111±0.077	3	0.74±0.06	0.077±0.055	0
4	10	0.76±0.09	0.074±0.039	4	0.74±0.06	0.091±0.034	2
8	1	0.73±0.11	0.130±0.142	0	0.74±0.06	0.072±0.055	0
8	5	0.80±0.08	0.075±0.059	3	0.74±0.06	0.081±0.042	0
8	10	0.79±0.11	0.069±0.063	1	0.75±0.06	0.073±0.041	1
16	1	0.68±0.17	0.112±0.108	0	0.74±0.05	0.081±0.027	0
16	5	0.77±0.11	0.115±0.061	0	0.75±0.07	0.097±0.043	0
16	10	0.76±0.11	0.102±0.047	0	0.74±0.07	0.079±0.052	0
∞	–	0.76±0.12	0.065±0.057	3	0.74±0.07	0.085±0.050	0

Table S4: Privacy: utility and fairness against the statistics budget ε and release period P, fairness budget $\tau = 0.005$. $\varepsilon = \infty$: statistics noise off, default period (P=1). mean±sd over 5 seeds. Fallback: number of seeds with no eligible feasible release after the warm-up, so the last model is reported. The ClipMean row is the unconstrained reference (no Lagrangian term).

ε	P	Bail AUC	Bail DPD	fallback
1	1	0.78±0.03	0.015±0.007	0
1	5	0.78±0.03	0.018±0.008	0
1	10	0.77±0.04	0.016±0.007	0
2	1	0.78±0.03	0.018±0.008	0
2	5	0.78±0.03	0.017±0.008	0
2	10	0.77±0.04	0.013±0.005	0
4	1	0.78±0.03	0.019±0.007	0
4	5	0.78±0.03	0.018±0.007	0
4	10	0.77±0.04	0.014±0.007	1
8	1	0.79±0.03	0.018±0.008	0
8	5	0.78±0.03	0.018±0.008	2
8	10	0.77±0.03	0.014±0.008	1
16	1	0.78±0.03	0.018±0.007	0
16	5	0.78±0.03	0.018±0.008	2
16	10	0.78±0.03	0.016±0.008	2
∞	–	0.78±0.03	0.015±0.007	2
ClipMean	–	0.78±0.03	0.019±0.009	–

Table S5: CLAMP ablations, one factor at a time around $C = 1$, $\eta = 0.5$, $\sigma_u = 0$, mean±sd over seeds. The last two rows at the default setting are the output-rule (last iterate) and direction-rule (proportional) ablations.

Dataset	Method	C	η	σ_u	AUC	DPD
Bail	CLAMP (last iterate)	1	0.5	0	0.78±0.05	0.023±0.004
Bail	CLAMP	0.5	0.5	0	0.78±0.04	0.019±0.004
Bail	CLAMP	1	0.1	0	0.78±0.04	0.023±0.004
Bail	CLAMP	1	0.5	0	0.78±0.04	0.023±0.004
Bail	CLAMP	1	0.5	0.01	0.78±0.05	0.024±0.004
Bail	CLAMP	1	0.5	0.05	0.78±0.05	0.022±0.004
Bail	CLAMP	1	2	0	0.78±0.05	0.023±0.004
Bail	CLAMP	2	0.5	0	0.78±0.04	0.024±0.004
Bail	CLAMP-P (proportional direction)	1	0.5	0	0.78±0.05	0.024±0.003
Credit	CLAMP (last iterate)	1	0.5	0	0.76±0.01	0.020±0.027
Credit	CLAMP	0.5	0.5	0	0.76±0.01	0.019±0.011
Credit	CLAMP	1	0.1	0	0.76±0.01	0.030±0.005
Credit	CLAMP	1	0.5	0	0.76±0.01	0.005±0.003
Credit	CLAMP	1	0.5	0.01	0.76±0.01	0.020±0.020
Credit	CLAMP	1	0.5	0.05	0.76±0.01	0.029±0.016
Credit	CLAMP	1	2	0	0.75±0.01	0.031±0.021
Credit	CLAMP	2	0.5	0	0.75±0.01	0.032±0.016
Credit	CLAMP-P (proportional direction)	1	0.5	0	0.76±0.01	0.007±0.004